Matrix inequalities and norms

Constant one in the logarithmic commutator inequality

Difficulty: challenging **Importance:** broadly interesting

Status: Solved

Historical rating rationale: Improving the known universal coefficient to its proposed sharp value is challenging; connections between matrix logarithms, entropy and quantum dynamics make the question broadly interesting.

Resolution: affirmative, 12 September 2026

Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, proves the exact inequality for every dimension and every complex positive definite pair in the original statement, with optimality already in order two. See **Theorem 1.1, Sections 2–5**, of the complete manuscript (Markdown, LaTeX).

A separate Codex AI-agent audit passed the full proof and checked its published relative-entropy input. This is informal review; no external human peer review, Lean verification or priority claim is asserted. Provenance and verified affiliation disclose AI assistance and credit PR #186, whose partial results left the universal upper bound open. The original target below is retained; ratings above are historical.

Problem statement

For every integer $n \geq 1$ and positive definite $A, B \in \mathbb{C}^{n\times n}$ with $\operatorname{tr}(A+B) = 1$, set $a = \operatorname{tr} A$, $b = \operatorname{tr} B$. Is

$$\|B\log(A+B) - \log(A+B)B\|_1 \leq -a\log a - b\log b$$

always true? The logarithm is the natural logarithm applied by spectral functional calculus; $\|X\|_1 = \operatorname{tr}(X^*X)^{1/2}$. Thus $a, b > 0$ and $a + b = 1$, and every expression is well-defined.

The target is specifically the coefficient 1 on the right. Existence of some dimension-independent coefficient has been proved and is not the open question. The coefficient-one question is explicitly suggested immediately after the original displayed conjecture.

This is a sharp matrix-function estimate for the rate at which noncommuting positive matrices mix. It connects perturbation estimates for the matrix logarithm with entropy production and quantum information.

References

1. K. M. R. Audenaert and F. Kittaneh, *Problems and Conjectures in Matrix and Operator Inequalities*, arXiv:1201.5232v3 (2012), §6, Conjecture 4, equation (26), and the sentence proposing $c = 1$. Full text.
2. K. M. R. Audenaert, *Quantum Skew Divergence*, J. Math. Phys. 55 (2014), 112202, §8, proof of small incremental mixing with coefficient 2. Full text.
3. A. Vershynina, *Entanglement rates for Renyi, Tsallis and other entropies*, arXiv:1803.07117v3 (2021), §8, conclusion, pp. 15–16, explicitly distinguishing the proved coefficient 2 and proposed coefficient 1. Full preprint.
4. Q. Ning, F.-Z. Guo, J. Zhang and Q.-Y. Wen, *On Bounding Entangling Rates and Mixing Rates in Some Special Cases*, Int. J. Theor. Phys. 55 (2016), 1686–1694. Published abstract, which reports coefficient one under additional restrictions; the subscription body was not independently inspected in this audit.

Historical status check (2026-09-10; superseded by the resolution above): Audenaert's theorem settles the existence claim with 2. Vershynina's 2021 revision distinguishes the proved coefficient from the proposed coefficient 1. Searches for "small incremental mixing sharp constant", "logarithmic commutator Audenaert constant one", and 2025/2026 found later special-case estimates but no announced universal coefficient-one proof or counterexample. This limited search does not certify that the conjecture remains open.